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freedom

THE WORLD'S MOST INNOVATIVE
ERGONOMIC MECHANICAL
KEYBOARD

v.210417

reddot winner 2020

c|net **Best keyboards 2021**

GOOD DESIGN AWARD®
GOLD WINNER

About the Zergotech Freedom

Welcome to the Zergotech Freedom,

Current ergonomic keyboards are either too intimidating to use, do not adequately address ergonomic issues, or both. We made it our mission to create a premium ergonomic mechanical keyboard that retains a familiar look and feel, while simultaneously integrating all possible ergonomic solutions. In our journey we conducted many trials and prototypes to deliver the most innovative ergonomic solutions. We based our design on the following principles:

1. Eliminating the twisting of hands by mobilising the wrists
2. Ensuring symmetry of reach for finger travel
3. Providing more even distribution of typing effort on each hand
4. Maximising other known ergonomic features
5. Minimizing, as much as possible, changes to the familiar keyboard layout

The result is an innovative keyboard that incorporates features such as:

1. Sliding palm rests
2. A unique semi-orthogonal layout
3. A split design
4. Programmable customizations
5. Multiple keyboard positions
6. Many other ergonomic considerations

This booklet outlines how the Zergotech Freedom delivers the optimal solution based on these design principles.

The Freedom has resolved RSI in many users and we are confident it will deliver similar results for you!

Safe typing!

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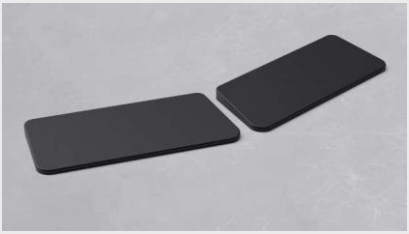
Innovative Palm Rests

Sliding Palm Rests

Our patented sliding palm rests are the most innovative and ergonomically effective feature of the Freedom. Wrist pain is the most common ergonomic complaint with standard keyboards. Typists that keep palms rested on a fixed surface while typing will have to stretch and twist their hands to reach their desired keys. Sliding palm rests allow you to slide over to any key while keeping your hands and forearms aligned. The sliding palm rests are given plenty of room to roam in a dedicated slider bay made of anti-scratch tempered glass for the lowest friction coefficient effect to give super smooth gliding.

Interchangeable Fixed Palm Rests

Discounting the sliding palm rests, the Zergotech Freedom is ergonomically effective in its own right. For users who prefer a more traditional typing experience, the Freedom provides interchangeable fixed palm rests. These can be swapped with the sliding palm rests at any time. The extra large palm rests accommodate hands of all sizes. All palm rests are made of a premium gel base with a synthetic leather covering.



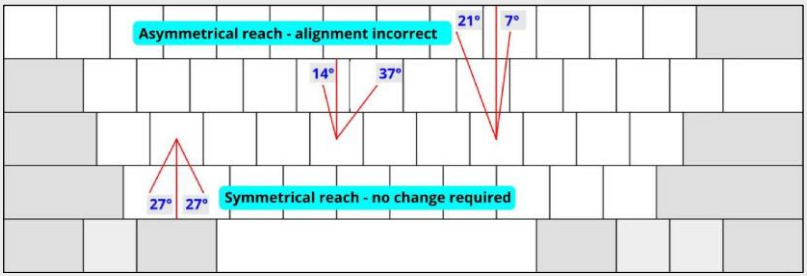
Semi-orthogonal Key Layout

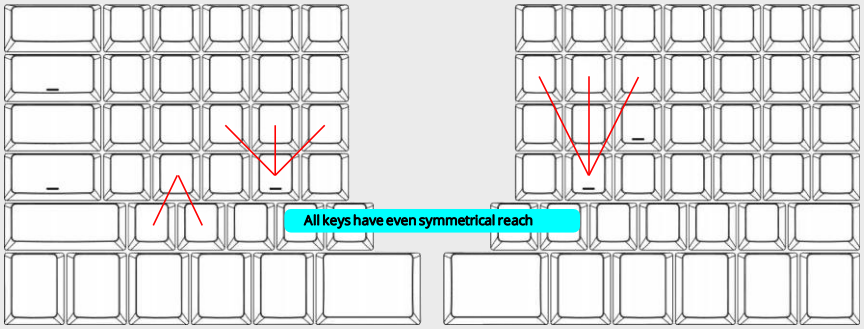
Behind the design

In keeping with the design philosophy of minimizing changes to the standard layout, the Freedom only changes what is necessary. Staggered columns are not required as the sliding palm rests mobilise the hands ensuring ease of reach

The semi-orthogonal layout retains the usual positions of keys on the home ASDF and the Shift ZXCV rows. The upper QWER is pushed a touch to the right to even the asymmetrical finger reach. Finally the locations of the standard modifier keys are unchanged.

The most effective changes to the layout are the three most frequently used, but poorly placed keys. The Enter, Backspace and Delete keys have been re-mapped from the extreme right on the standard keyboard, to the left hand. This both reduces the amount of twisting and travel to the keys while giving the left hand more typing work, more evenly distributing typing between the left and right hands.





Multiple Ergonomic Angles

Default, Tented, Incline, and Decline

Very few ergonomic keyboards are able to achieve multiple positions. Those that do, require complex and unattractive apparatus. It was a geometric challenge to achieve these multiple positions while keeping a comfortable 5° tenting in the standard, inclined, and declined positions.

1. Default Position
The Freedom ergonomic keyboard has a default tented angle of 5° away from centre. Many users find this angle is enough to relieve pain.

2. Increased Tented Position
For users seeking more impactful relief from pronation stress, the Freedom can be tented to a more aggressive 12°. This position fully aligns the keyboard with your forearms.

3. Declined Position
Users who sit at a relatively higher position to their desk may prefer a declined position. The declined position still retains the default 5° tented angle, combining decline with tenting.

4. Inclined Position
Those who sit relatively lower than their desk may prefer an inclined position. The inclined position also retains the default 5° tented angle, combining incline with tenting.

ENTER
SPACE
FN SHIFT
NUM LOCK
FN LOCK
CAPS LOCK
SHIFT
CTRL

Multiple Layouts

Three key layers

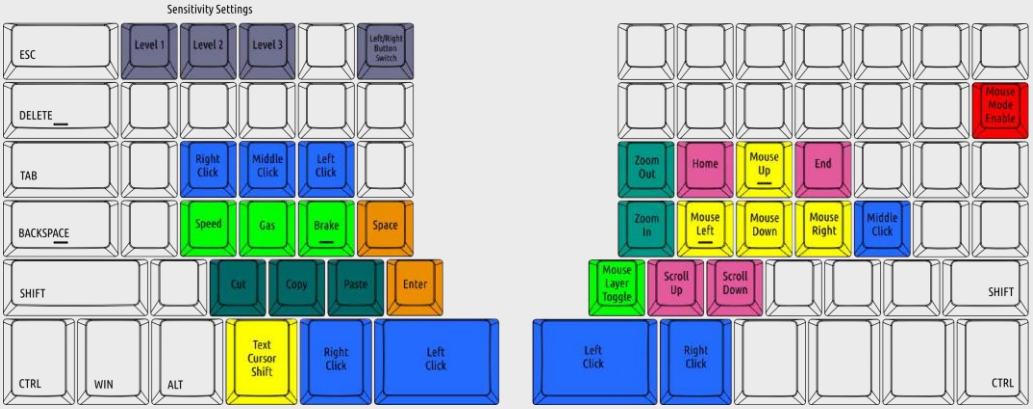
The Freedom has three layers of integrated key layouts. Standard dedicated navigation and numerical keys not only require a physical movement of the arm, it also takes up a considerable amount of space to the right. Adding a right handed mouse causes a disproportionate reliance on the right half of a desktop. This creates a heavy right lean on the right arm, contributing to neck and shoulder pain. Integrating the navigation and numpad layers into the main board relieves this pain and increases efficiency by giving your fingers immediate access to these keys.

The Default layout	Printed in White, this layer has standard characters for typing
The Function layout	Denoted by Blue key prints, the Function layout incorporates the navigation keys and useful shortcuts which are activated by either a FN Shift or FN Lock key
The NUM layout	Denoted by Yellow key prints, the NUM layout incorporates the numerical keys and the backquote and tilde keys, activated by either a NUM Shift or NUM Lock Key

Embedded Ergo Mouse

The Freedom also comes with an ergonomically effective feature - an embedded mouse controlled by the keyboard itself. No longer do you need to stretch your arm and reach for a physical pointing device, the cursor is now at your fingertips.
With intuitive and seamless navigation, the embedded mouse provides 12 speed settings, common key shortcuts, zoom controls and all general navigation buttons.
Features include:

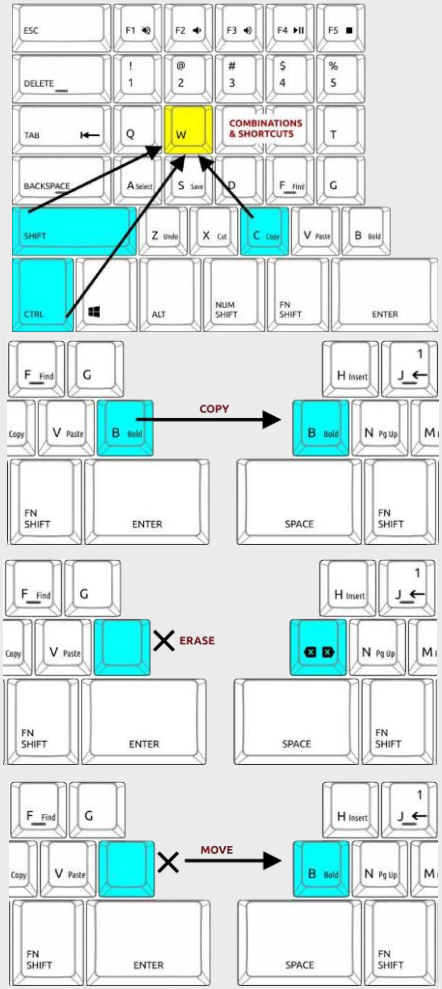
- 12 Speed Settings
- Intuitive Handing and Navgiation
- Mouse Pointer / Text Cursor Shift
- Fingertip Control
- Common Shortcuts
- Zoom Controls
- Standard Modifier Keys

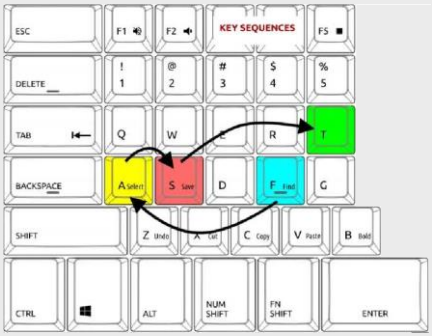


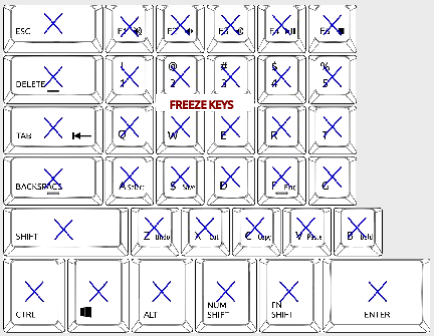
KeyKortex Programming Module

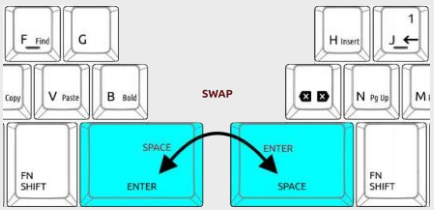
Programmable

The Freedom comes with the powerful KEYKORTEX on-board programming module. KEYKORTEX is the most advanced on-board programming module found in any keyboard on the market. The Freedom's programming module gives you the flexibility to create shortcuts, alter layouts and remap keys, freeze keys, create powerful combinations and sequences on the fly and with no added software.









Multiple Configurations

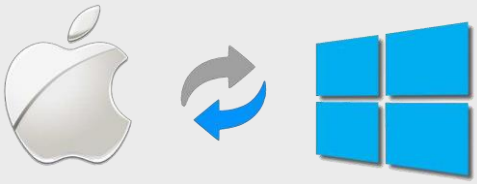
Multi Lingual and Dual Standard

The Freedom supports both ANSI and ISO standards, providing compatible layouts and keyprints for:

- US QWERTY ANSI Layout
  (also used in Australia and New Zealand)
- UK QWERTY ISO Layouts
- German QWERTZ ISO Layout
- French AZERTY ISO Layout

Multi Operating Systems

Switch between MAC OS and Windows PC keyboard layouts on the fly. The Freedom is also Linux compatible.
The Freedom is configurable for each system by adapting its modifier keys to match the respective OS, and changing the in-built default shortcuts to be compatible with the enabled operating system.



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ESC
DELETE
TAB
BACKSPACE
SHIFT
CTRL
ALT
NUM SHIFT
FN SHIFT
ENTER
SPACE
NUM LOCK
FN LOCK
CAPS LOCK

Mechanical Switches

Why Mechanical Key Switches?

Standard plastic keyboards make use of cheap rubber domed keys. These keys have very low travel to actuation, heavy key presses, short lifespan and a very unpleasant typing experience. Mechanical switches are manufactured for precision. They are made of high quality internal pieces for a superior typing experience. They feature further key travel, accurate and consistent actuation force, tactile feedback, and a long life space beyond 80 million keystrokes.

Kailh Box Switches

Kailh produced switches are among the market leaders in mechanical keyboard switches. Renowned for their quality and longevity, Kailh has been the choice of industry leading device manufacturers, including Samsung, Microsoft, HP and more. Zergotech has chosen the Kailh's patented Box switch which provides more stability with each keystroke, removing an unpleasant wobble, characteristic of unbounded single stem switches.

The Freedom has the option of either a Brown Box Tactile Switch, or a White Box Clicky Switch.

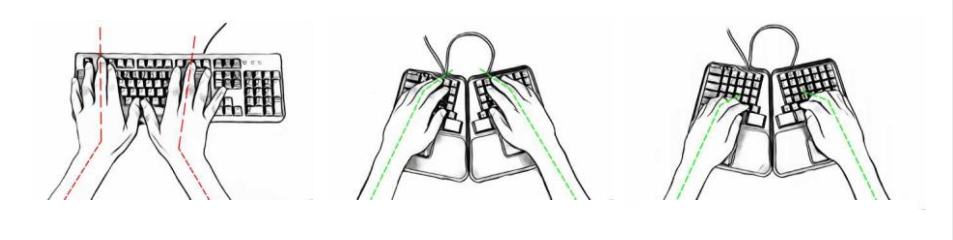
The brown box switch has tactile feedback without audible clicks making it more suitable for quiet office environments.

The white box switch has a satisfying audible click for expert level feedback and control.

Ergonomic Solutions

Solving Vertical Reach

The main feature of the Freedom keyboard is the patented Sliding Palm Rests. The ability to roam means the more distant keys are no longer out of reach. The ability to slide down to the lower row also removes the need to compress your fingers.



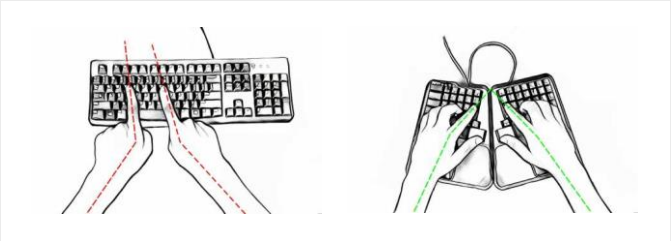
Solving Horizontal Deviation

Repeated horizontal deviations and extended periods of over-reach place excessive strain on the wrist - the largest source of typing RSI pain. The Freedom resolves this problem by keeping the wrists aligned using the sliding palm rests. The Freedom also remaps the extreme right keys - Enter, Backspace and Delete - to the left board. By integrating the navigation keys and numerical pad into the main board, your hands are never required to leave their resting position.

Ergonomic Solutions

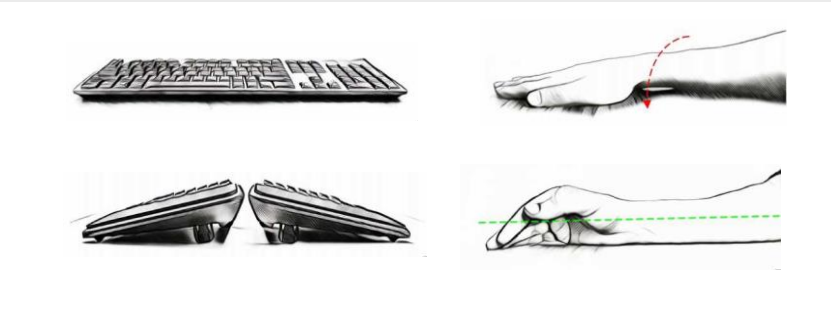
Solving Asymmetry

Standard issue flat keyboards have asymmetrical and unevenly offset top rows. This creates unequal travel, and either an under-reach or over-reach of corresponding fingers. The Freedom solves this problem with the semi-orthogonal design by aligning only the top offset rows respective to the home row. The bottom row is unchanged as it is already offset by an equal amount to the home row.



Solving Pronation

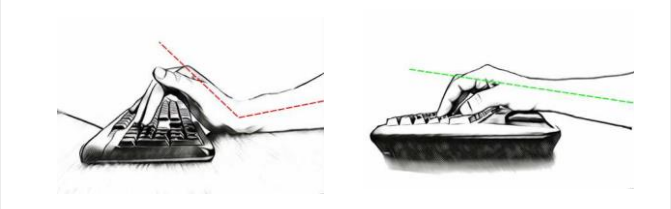
Forearm pronation is required to operate a flat standard keyboard, exerting most of its pressure at the wrist. The Freedom is never completely flat, with it's default position having a small slant giving the keyboard a natural feel. There Freedom also has the option of extending the feet for an increased tented angle.



Ergonomic Solutions

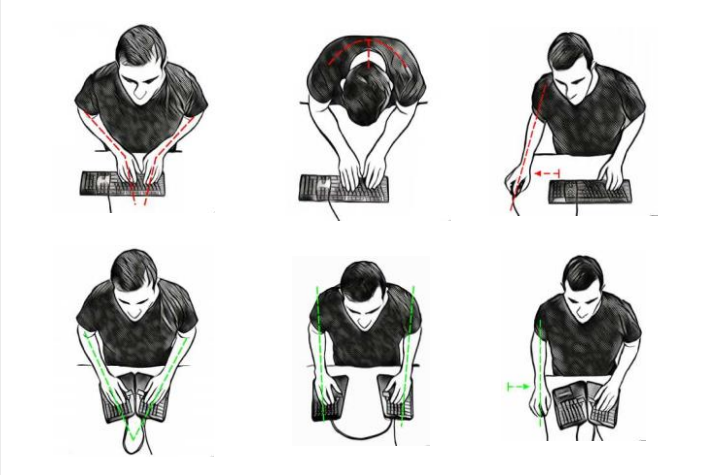
Solving Wrist Extension

Wrist extension for prolonged periods contributes to fatigue, wear, and inflammation of the wrists. The Freedom's sliding and fixed palm rests resolve this problem by raising the wrists relative to the keys.



Solving Slouching and Right Lean

A standard keyboard close to your body results in strained wrists. Pushed further away, wrist alignment is slightly improved but causes slouching. A wide layout and right handed mouse usage puts heavy reliance on the right arm, resulting in right shoulder lean causing pain. The Freedom is split and rotated inwards to allow close body typing, giving you correct posture in any position.



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PATENT PENDING

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FREEDOM OF MOVEMENT

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